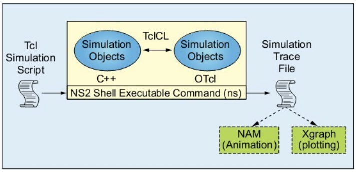

Figure 1: Basic architecture of ns-2 *(Courtesy: tutorialsweb.com)*

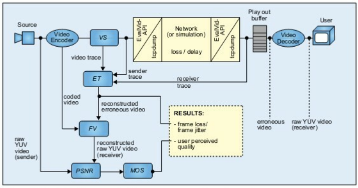

Figure 2: EvalVid framework *(Courtesy: tutorialsweb.com)*

calculation. It can regenerate received video, which can be compared with the originally transmitted video in terms of many other useful metrics like video quality measure (VQM), structural similarity (SSIM), mean opinion score (MOS), etc. The framework of EvalVid is shown in Figure 2.

Integration of ns-2 and EvalVid

Raw video can be encoded by the required codec using tools provided by EvalVid. Video traces that contain information about video frames, like the frame type, time stamp, etc, are generated by using EvalVid. These trace files are used as a traffic source in ns-2. The integrated architecture of ns-2 and EvalVid is shown in Figure 3.

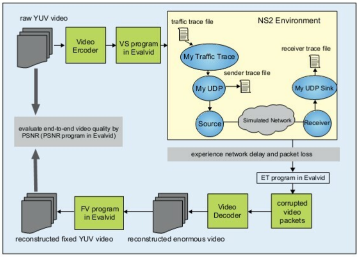

Figure 3: Integrated architecture of ns-2 and EvalVid *(Courtesy: csie.nqu.edu.tw)*

Scripting an agent in ns-2

Agents created in ns-2 need to read the generated video trace file. This agent code is easily available on the Internet. The following steps must be taken for integrating the agent code in ns-2.

Step 1: Put a *frametype_* and *sendtime_ field* in the *hdr_cmn* header. The *frametype_ field* is to indicate which frame type the packet belongs to. The 'I' frame type is defined to 1, 'P' is defined to 2, and 'B' is defined to 3. The *sendtime_ field* records the time the packet was sent. It can be used to measure end-to-end delay.

You can modify the file *packet.h* in the common folder as follows:

```
struct hdr_cmn {
  enum dir_t { DOWN= -1, NONE= 0, UP= 1 };
  packet_t ptype_;       // packet type (see above)
  int     size_;             // simulated packet size
  int     uid_;              // unique id
  int     error_;             // error flag
  int     errbitcnt_;      //
  int     fecsize_;
  double      ts_;           // timestamp: for q-delay
measurement
  int     iface_;             // receiving interface (label)
  dir_t direction_;       // direction: 0=none, 1=up,
-1=down
  // source routing
  char src_rt_valid;
  double ts_arr_; // Required by Marker of JOBS
 //add the following three lines
  int frametype_;              // frame type for MPEG video
transmission
  double sendtime_;    // send time
unsigned long int frame_pkt_id_;
```

Step 2: You can modify the file *agent.h* in the common folder as follows:

```
class Agent : public Connector {
 public:
  Agent(packet_t pktType);
  virtual ~Agent();
  void recv(Packet*, Handler*);
 ………..
inline packet_t get_pkttype() { return type_; }
// add the following two lines
  inline void set_frametype(int type) { frametype_ = type; }
  inline void set_prio(int prio) { prio_ = prio; }

 protected:
  int command(int argc, const char*const*argv);
  ………..
  int defttl_;       // default ttl for outgoing pkts
                   // add the following line
  int frametype_;                    // frame type for
MPEG video transmission
```